

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2011

BIOLOGY

9648/03

Higher 2

23 August 2011
2 hours

PAPER 3 Applications Paper

Additional Material: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and candidate number in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Answer **all** questions.

The intended number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten your work securely together with the string provided.

You are reminded of the need for good English.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	72

This document consists of **12** printed pages.

- 1 (a) (i) Plasmids used for cloning of foreign genes in bacteria usually have at least two marker genes. On the diagram of a plasmid shown in Fig. 1.1, draw two named marker genes and the other components that are required for the plasmid to be suitable for use in the cloning of foreign genes. You are to label these components with specific names where applicable. [2]

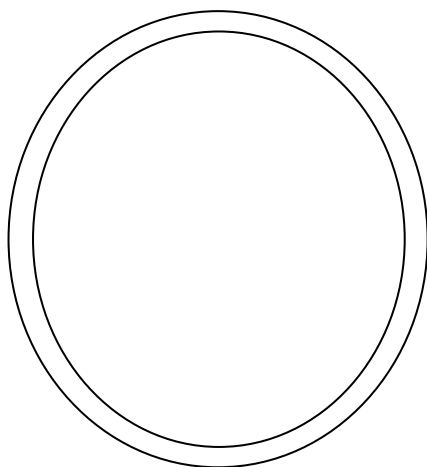


Fig. 1.1

- (ii) Explain the advantage of having two marker genes instead of one in the plasmid mentioned above.

[2]

- (iii) The process of cloning eukaryotic genes in plasmids is more complicated than that of prokaryotic genes. Briefly explain why this is so.

[1]

- (b) (i) One application of gene cloning is the construction of genomic libraries. Outline the process of constructing a genomic library.

[3]

- (ii) Another library that can be constructed is the cDNA library. State how this library differs from the genomic library constructed for an organism.

[2]

- (iii) Explain how bacterial colonies containing a gene of interest could be identified in the cDNA library.

[3]

[Total: 13]

- 2 One approach of gene therapy to treat cystic fibrosis uses viruses to deliver normal alleles of the CFTR gene into epithelial cells of the airways.

A team of researchers in the USA developed a new strain of non-pathogenic adeno-associated virus (AAV), AAV2.5T. Genes for the CFTR protein and the enzyme luciferase were inserted into the DNA of the viruses. Luciferase catalyses the production of a green fluorescent protein when luciferin is added.

The normal AAV strain and the AAV2.5T strain were added to cultures of epithelial cells from the airways. After adding luciferin, the number of cells that had taken up the viral genes was estimated using the intensity of the green fluorescence which developed. The results are shown in Fig. 2.1.

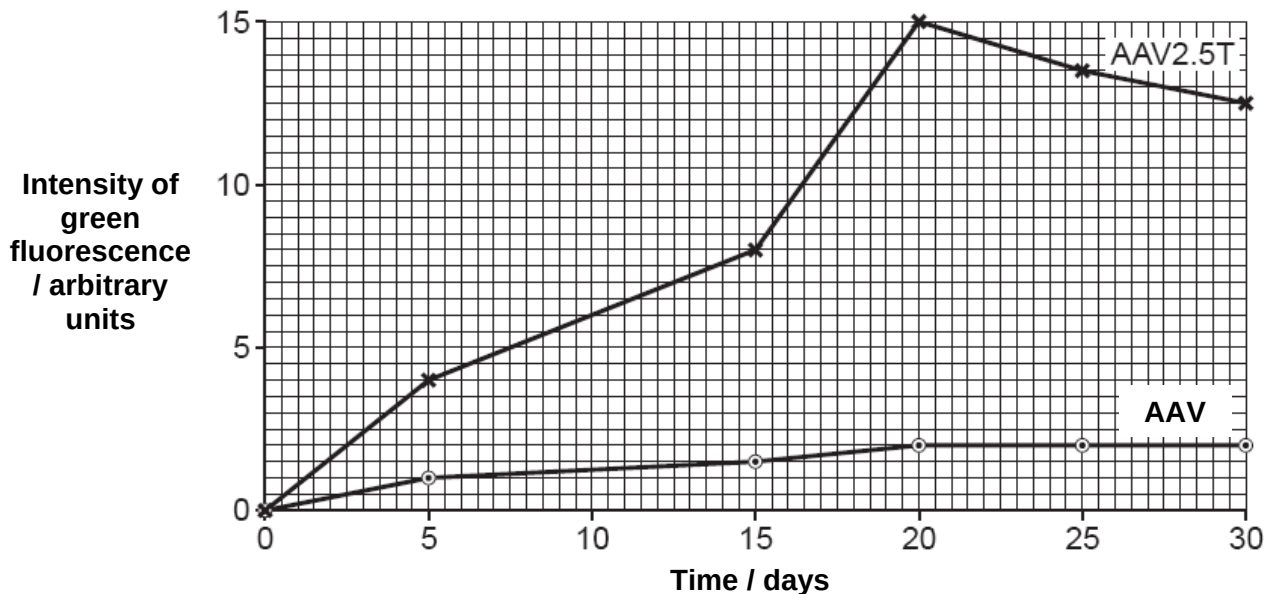


Fig. 2.1

- (a) Explain why the researchers inserted a gene for luciferase into the viral DNA.

.....
 [1]

- (b) With reference to Fig. 2.1, compare the ability of the two viral strains, AAV and AAV2.5T, to infect epithelial cells of the airways.

.....

 [3]

- (c) Suggest why a decrease in intensity of green fluorescence was detected in cells infected with AAV2.5T during the last 10 days.

[2]

- (d) Describe how delivering normal alleles of the CFTR gene into epithelial cells in the airways could relieve the symptoms of cystic fibrosis.

[4]

- (e) In some cystic fibrosis patients, the allele has a single-base mutation which produces a 'nonsense' (stop) codon within the gene. Explain how this mutation affects the production of normal CFTR protein.

[2]

[Total:12]

- 3 Atlantic salmon (*Salmo solar*) is a salmon of great commercial value in the food industry. It can be marketed as whole or processed; sliced, filleted, smoked, canned or marinated. The popularity of Atlantic salmon has led to its substitution with other non-salmon species or with salmon species of lower nutritional value being marketed and labelled as “Atlantic salmon” nevertheless.

In a validation exercise to identify and distinguish Atlantic salmon from other fish species, genus-specific PCR method was used. In this method, genetic sequence of *ITS1* gene from Atlantic salmon was obtained and was used to design primers and a probe (radioactive) specific to salmon from the *Salmo* genus. Fig. 3.1 below shows the *ITS1* gene as well as the binding positions of the primers and the probe.

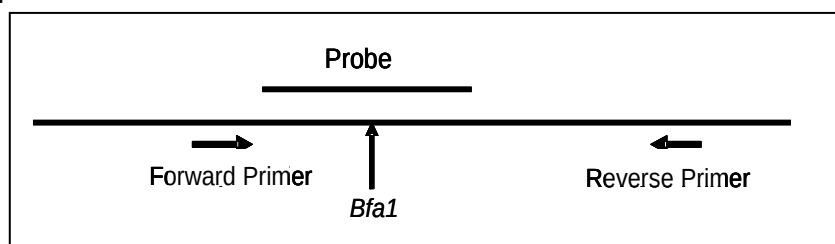


Fig. 3.1

“Atlantic salmon” labelled products were bought from various markets and their DNA were extracted. The extracted DNA was amplified via PCR using the primers shown in Fig. 3.1. Gel electrophoresis and Southern blotting were carried out on the resulting amplified products to obtain the results in Fig. 3.2 below.

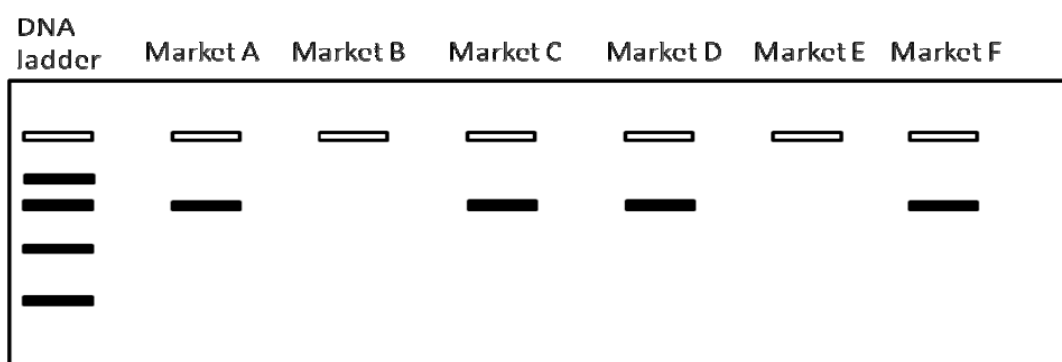


Fig. 3.2

To confirm the identity of the bands shown in Fig. 3.2, the amplified DNA from the various samples were treated with restriction enzyme *Bfa1*. The position of the *Bfa1* restriction site is shown in Fig. 3.1. The radioactive probe in Fig. 3.1 was used to visualize the band patterns shown in Fig. 3.3.

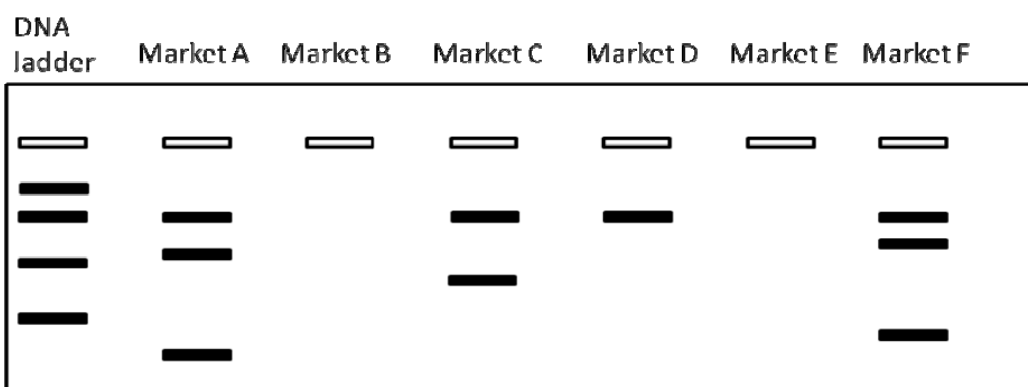


Fig. 3.3

(a) (i) Explain why the two types of primers are needed to amplify the DNA segment.

[1]

(ii) Briefly describe the process of how Fig. 3.3 was obtained after the amplified DNA from the various samples were digested by *Bfa*1.

[3]

(iii) Using information from Fig. 3.1 and Fig. 3.2, identify and explain fully which market(s) does not sell fish from the *Salmo* genus.

[2]

(iv) Using information from Fig. 3.3, identify and explain fully which market(s) sells genuine Atlantic salmon.

[2]

(v) Suggest how Fig. 3.3 can be used to infer the evolutionary relationship between different species of salmon.

[2]

Transgenic Atlantic salmon contains two foreign genes inserted into their genome. Fig. 3.4 shows the gene construct of the two genes inserted into the Atlantic salmon; a promoter sequence from Ocean Pout and a growth hormone gene from the Chinook salmon. Some companies chose not to label their transgenic Atlantic salmon product despite regulations in food labelling for the fear of a drop in sales. Another investigation was carried out to detect the presence of non-labelled transgenic salmon being sold in the market.

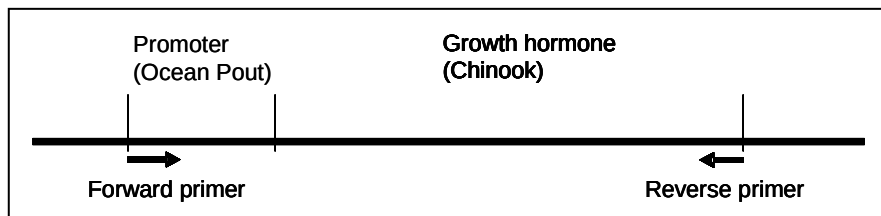


Fig. 3.4

- (b) (i)** Draw on Fig. 3.4 the position of the probe that would enable one to distinguish between wild type Atlantic salmon, transgenic salmon, Ocean Pout and Chinook salmon. [1]

PCR was carried out on all the samples to be investigated using the primers as shown in Fig. 3.4.

- (ii)** Demonstrate how your probe proposed in **(b)(i)** would distinguish the transgenic Atlantic salmon from the wild type, Ocean Pout and the Chinook salmon by drawing on Fig. 3.5 their respective band positions. [2]

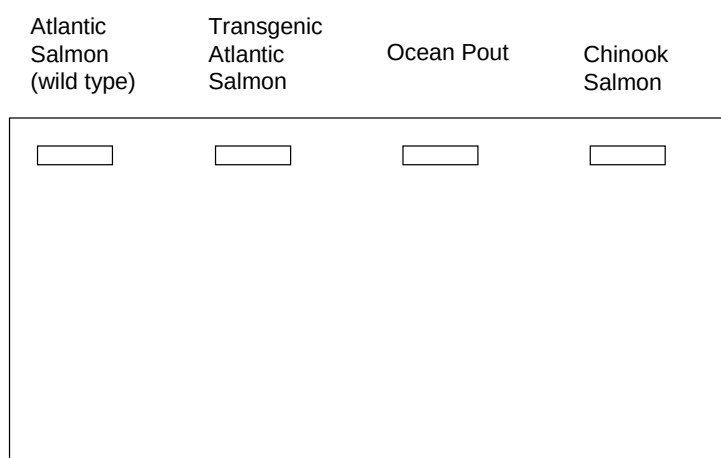


Fig. 3.5

- (iii)** Explain how the two inserted genes enable the transgenic salmon to meet the world's demand for food.

[2]

[Total: 15]

Planning Question**4 Title: Effect of different light intensities on the rate of photosynthesis in algae**

You are required to plan, but not carry out, an investigation of the effect of different light intensities on the rate of photosynthesis in algae.

You are provided with unicellular photosynthetic algae which are trapped in alginate beads. When carbon dioxide in solution is absorbed by the algae for photosynthesis, the change in carbon dioxide level can be detected using hydrogen carbonate indicator. The indicator is red when equilibrated with atmospheric carbon dioxide. It changes to yellow when more carbon dioxide is added and changes to purple when carbon dioxide is removed.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **MUST** use.

Apparatus and Materials

- Algal-alginate beads
- Boiled and cooled algae trapped in alginate beads
- Hydrogen carbonate indicator
- A pair of forceps
- Small transparent containers
- Metre rule
- Light source (150 watt)
- Thick sheet of clear glass 150 mm × 100 mm with safe edges
- Cuvettes and spectrophotometer
- A variety of different sized beakers, measuring cylinders and syringes for measuring volumes

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure,
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures,
- an explanation of dependent and independent variables involved,
- relevant and clearly labeled diagrams,
- how you will record your results and ensure they are accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms.

[Total:12]

ACJC

[Turn over

Free-response question

Write your answers in the lined pages provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Patients who are deficient in alpha galactosidase suffer from Fabry disease. They can be treated by enzyme replacement therapy through regular injection of the enzyme into their bloodstream. Explain how alpha galactosidase can be mass produced using *E. coli* for the enzyme replacement therapy. [8]

(b) Discuss the social and ethical considerations for the use of gene therapy. [6]

(c) Explain the advantages and limitations of PCR. [6]

[Total:20]